KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 13

YEAR 9 MATHEMATICS

Similar figures

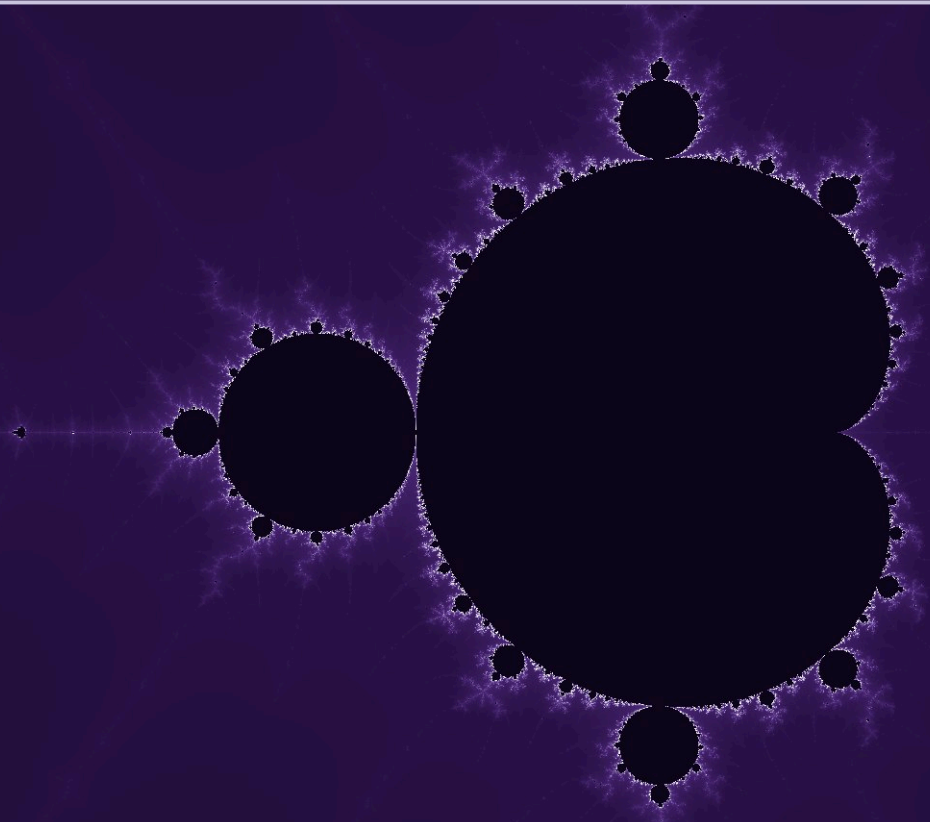
4 lessons · three pages each · Easy drill with diagrams, then Medium and worded problems

13.1 Similar figures

13.2 Enlargements

13.3 Finding unknown sides

13.4 Shadows and scale drawings



NAME _____ CLASS _____ TEACHER _____

Artwork: a small copy of the whole Mandelbrot set on its antenna (the set is self-similar), coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **three pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1 and 2 are **Easy**: small steps, one question at a time. Page 3 is **Medium**, then three **worded problems**.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Each question says what to do, step by step. Write each answer on the line that names it, like **Scale factor =**.

3. Some Medium questions come from earlier lessons. The tag (like **12.2** or **Year 8**) says which one.

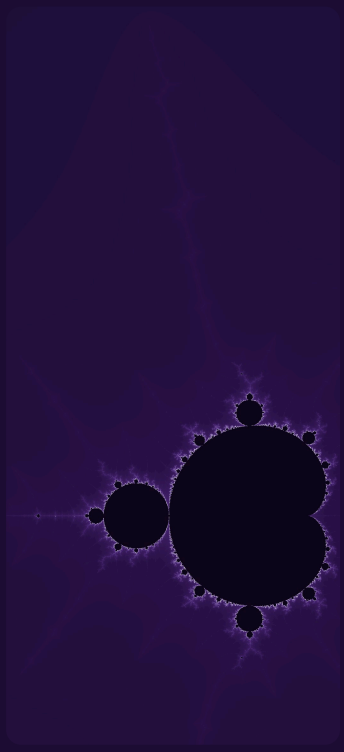
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Write scale factors as **whole numbers or fractions in simplest form**, and lengths with their **units**.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
13.1	Similar figures	3		☐
13.2	Enlargements	6		☐
13.3	Finding unknown sides	9		☐
13.4	Shadows and scale drawings	12		☐

In this chapter you will

- name the matching sides and angles of similar figures and write a similarity statement (ABC ||| PQR)
- find the scale factor of a pair of similar figures
- draw enlargements and reductions on a grid
- test whether two figures are similar
- find unknown sides of similar figures
- use shadows, scale drawings and maps to find real heights and lengths



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a small copy of the whole Mandelbrot set on its antenna (the set is self-similar), coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

13.1 Similar figures

EASY

Learning intention We are learning to name the matching parts of similar figures and find the scale factor.

Success criteria
- ☐ I can name the matching sides and angles of similar figures
- ☐ I can write a similarity statement with |||
- ☐ I can find the scale factor: image side ÷ original side

NOTES — Key ideas for this lesson

TEACHER COPY

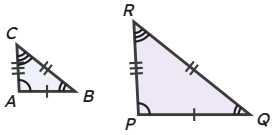
KEY TERMS

similar: the same shape, any size
matching: in the same place on both figures
image: the new figure
scale factor: how many times bigger
enlargement: scale factor more than 1
reduction: scale factor less than 1

FORMULAS

Scale factor = image side ÷ original side
ABC ||| PQR: A with P, B with Q, C with R
Matching angles are equal
Matching sides are in the same ratio

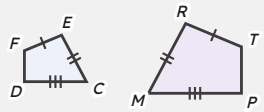
STEPS
1. match the vertices
2. pick a pair of matching sides
3. image side ÷ original side
4. simplify the fraction



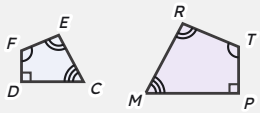
WE DO — With your teacher: copy what they write on the board.

Example 1 The quadrilaterals FECD and TRMP are similar.

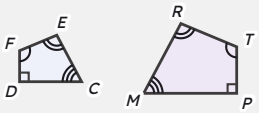
a List the matching sides.



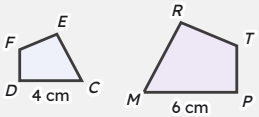
b List the matching angles.



c Write the similarity statement.



d Find the scale factor.



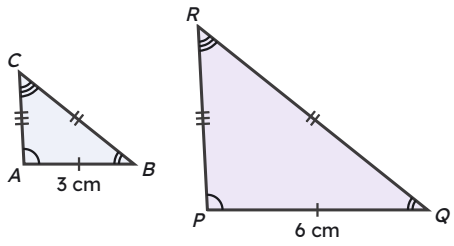
YOU DO — On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Matching sides have the same number of ticks · matching angles have the same number of arcs · simplify the scale factor

QUESTION 1 Name the matching side. Then scale factor = image side ÷ original side.

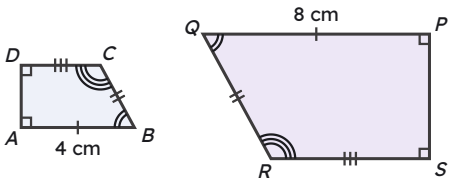
a



AB matches _____

Scale factor = _____

b



AB matches _____

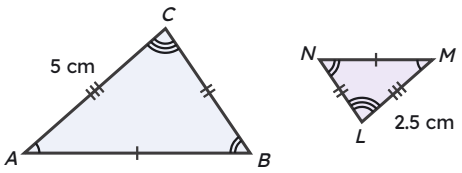
Scale factor = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Turn or flip the second figure in your head until it sits like the first · count the squares along each side · no marks: shortest side with shortest side

QUESTION 2 Name the matching side. Then scale factor = image side ÷ original side.

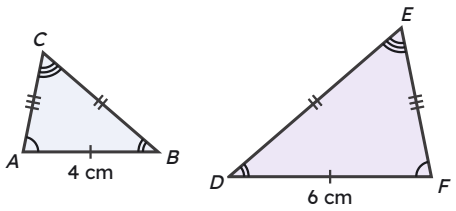
(a)



CA matches _____

Scale factor = _____

(b)

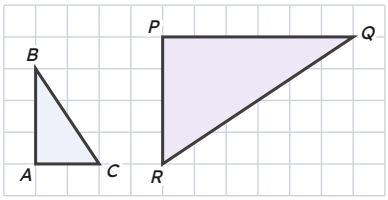


AB matches _____

Scale factor = _____

QUESTION 3 Count the squares along each side. Name the matching side, then the scale factor.

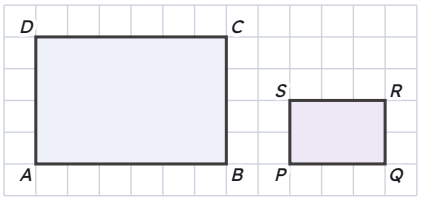
(a)



AB matches _____

Scale factor = _____

(b)

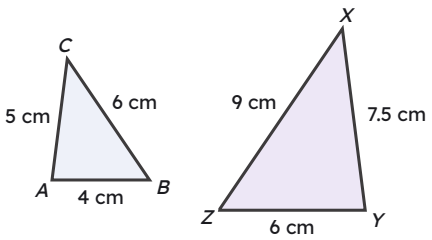


BC matches _____

Scale factor = _____

QUESTION 4 No marks: match shortest, middle and longest sides. Write the statement, then the scale factor.

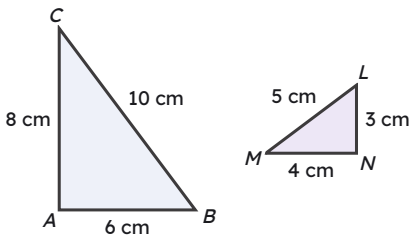
(a)



Statement _____

Scale factor = _____

(b)



Statement _____

Scale factor = _____

13.1 Similar figures

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Triangle ABC: AB = 6, BC = 9, CA = 12 cm. Triangle XYZ: XY = 8, YZ = 4, ZX = 6 cm.**

a List the matching sides, shortest to longest.

b Write the similarity statement.

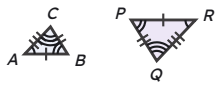
c Find the scale factor. Enlargement or reduction?

d Which angle matches ∠B?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Shortest with shortest, longest with longest · the statement lists matching vertices in the same order · less than 1: a reduction

5 Write the similarity statement.

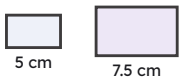


Statement _____

6 Simplify the ratio 12 : 18. Yr 8

Answer _____

7 Find the scale factor.



Scale factor = _____

8 PQR ||| XYZ: match side QR and ∠P.

QR matches _____

∠P matches _____

9 A die is rolled. Find P(less than 3). 12.1

Answer _____

10 Scale factor? Enlargement or reduction?



Scale factor = _____

Type _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Triangle ABC has sides AB = 6 cm, BC = 8 cm and CA = 10 cm. Triangle KLM has sides KL = 9 cm, LM = 12 cm and MK = 15 cm. Write the similarity statement and find the scale factor.

Answer _____

2 A drawing of a car is 12 cm long. The real car is 4.2 m long. Find the scale factor from the drawing to the real car. (Change 4.2 m to cm first.)

Answer _____

3 Are all squares similar? Are all rectangles similar? Explain, using a 1 cm by 2 cm rectangle and a 2 cm by 3 cm rectangle.

Answer _____

13.2 Enlargements

EASY

Learning intention We are learning to draw enlargements and reductions on a grid.

Success criteria
- ☐ I can count the lengths of a shape on a grid
- ☐ I can multiply every length by the scale factor
- ☐ I can draw the image of a shape on a grid

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

enlargement: every length × the same number
reduction: an enlargement that is smaller
scale factor: the number every length is × by
image: the shape after enlarging
grid square: one unit of length
angles: stay the same

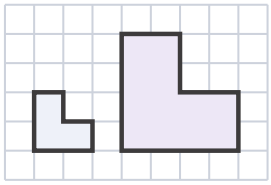
FORMULAS

Image length = original length × scale factor

Scale factor 2: every length doubles

Scale factor 1/2: every length halves

The angles do not change



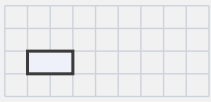
STEPS

1. count each side in squares
2. multiply by the scale factor
3. draw the image side by side
4. check the angles match

WE DO — With your teacher: copy what they write on the board.

Example 1 Write the lengths of each shape and of its image. Then draw the image.

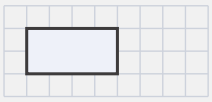
a Scale factor 2



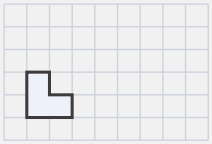
b Scale factor 3



c Scale factor 1/2



d Scale factor 2



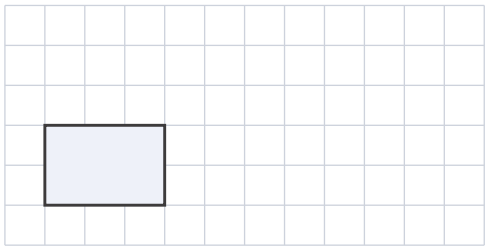
YOU DO — On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Image length = original length × scale factor · scale factor 1/2: halve every length · the angles stay the same

QUESTION 1 Count each side in squares. Multiply by the scale factor. Draw the image on the grid.

a Scale factor 2



Original lengths _____

Image lengths _____

b Scale factor 3



Original lengths _____

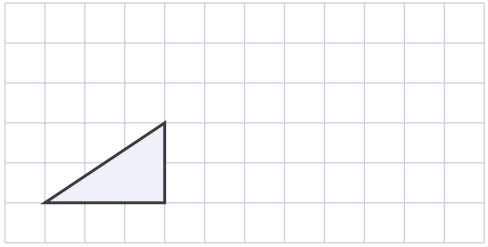
Image lengths _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Count every side, going round · multiply each one by the scale factor · draw the image in the empty part of the grid

QUESTION 2 Count each side in squares. Multiply by the scale factor. Draw the image on the grid.

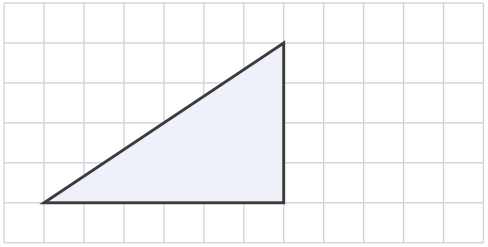
(a) Scale factor 2



Original lengths _____

Image lengths _____

(b) Scale factor 1/2

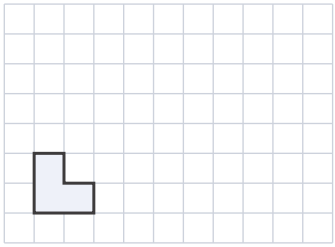


Original lengths _____

Image lengths _____

QUESTION 3 Count each side in squares. Multiply by the scale factor. Draw the image on the grid.

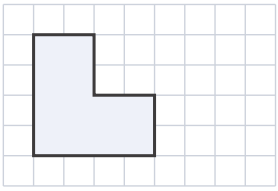
(a) Scale factor 3



Original lengths _____

Image lengths _____

(b) Scale factor 1/2



Original lengths _____

Image lengths _____

QUESTION 4 Halve every length. Write the image's lengths, then draw the image on the grid.

(a) Scale factor 1/2

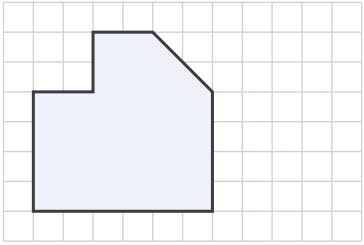


Image lengths _____

(b) Scale factor 1/2

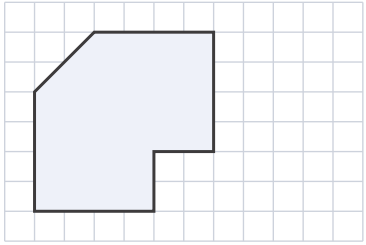


Image lengths _____

13.2 Enlargements

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **A rectangle is 4 cm long and 6 cm wide. It is enlarged.**

a Find the image's lengths for scale factor 3.

b Find the image's lengths for scale factor 1/2.

c Find the image's lengths for scale factor 2.5.

d Another image is 6 cm by 9 cm. Find its scale factor.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Image length = original length × scale factor · scale factor less than 1: a reduction · scale factor = image ÷ original

5 Scale factor 3: write the image lengths.



Image lengths _____

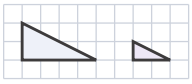
6 Divide \$40 in the ratio 3 : 5. Yr 8

Answer _____

7 A 4 cm by 6 cm photo: scale factor 2.5.

Image lengths _____

8 Find the scale factor.



Scale factor = _____

9 A coin is tossed twice. Find P(two heads). 12.4

Answer _____

10 Sides 3, 4, 5 cm; scale factor 1/2.

Image sides _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A photo 12 cm long and 8 cm wide is enlarged by a scale factor of 3. What are the length and width of the enlarged photo?

Answer _____

2 A model car is a reduction of a real car with scale factor 1/20. The real car is 4.2 m long. How long is the model, in cm?

Answer _____

3 A square with sides of 3 cm is enlarged by a scale factor of 2. Draw both squares. How many times larger is the area of the image?

Answer _____

13.3 Finding unknown sides

EASY

Learning intention We are learning to find unknown sides of similar figures and test whether figures are similar.

Success criteria
- ☐ I can find the scale factor from a pair of matching sides
- ☐ I can find an unknown side of similar figures
- ☐ I can test whether two figures are similar

NOTES — Key ideas for this lesson

TEACHER COPY

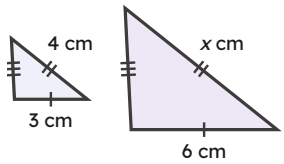
KEY TERMS

similar: matching sides in the same ratio
unknown side: the side marked x
scale factor: image side ÷ original side
matching sides: same ticks, or same order
ratio: one length ÷ another
test: are all the ratios equal?

FORMULAS

Scale factor = image ÷ original
Image side = original × scale factor
Original side = image ÷ scale factor
Similar: all the ratios are equal

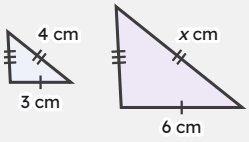
STEPS
1. find a measured matching pair
2. scale factor = image ÷ original
3. find the side matching x
4. × or ÷ by the scale factor



WE DO — With your teacher: copy what they write on the board.

Example 1 Each pair of figures is similar. Find the scale factor, then x.

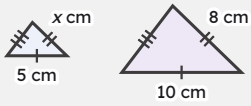
a x on the image



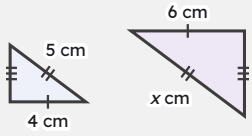
b Rectangles



c x on the original



d A fraction scale factor



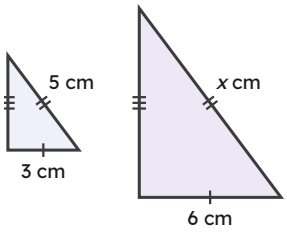
YOU DO — On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Scale factor = image side ÷ original side · x on the image: × the scale factor · x on the original: ÷ the scale factor

QUESTION 1 Scale factor = image side ÷ original side. Then find x.

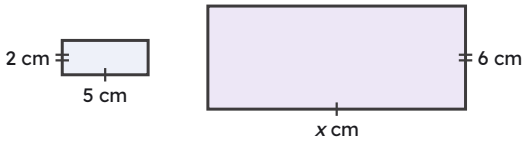
a



Scale factor = _____

x = _____

b



Scale factor = _____

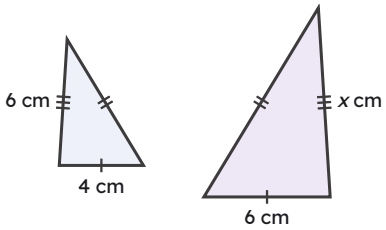
x = _____

YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Use the pair of matching sides with both lengths for the scale factor · then find the side that matches x · check: is x bigger or smaller?

QUESTION 2 Scale factor = image side ÷ original side. Then find x.

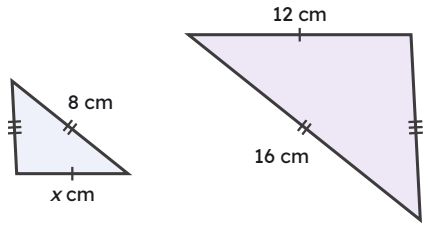
(a)



Scale factor = _____

x = _____

(b)

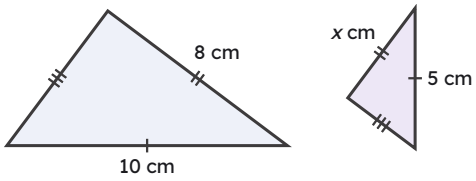


Scale factor = _____

x = _____

QUESTION 3 Scale factor = image side ÷ original side. Then find x.

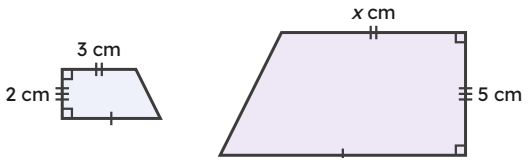
(a)



Scale factor = _____

x = _____

(b)

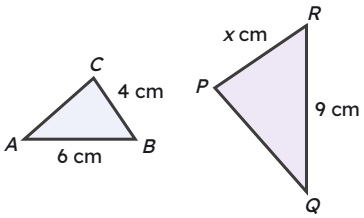


Scale factor = _____

x = _____

QUESTION 4 The sides are not marked: use the statement to match them. Find the scale factor, then x.

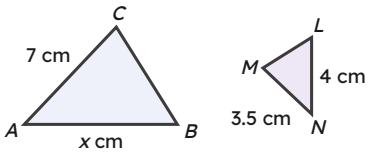
(a) ABC ||| QRP



Scale factor = _____

x = _____

(b) ABC ||| NLM



Scale factor = _____

x = _____

13.3 Finding unknown sides

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Test whether each pair is similar, then find the unknown sides.**

a Rectangles 20 mm by 12 mm and 29 mm by 18 mm.

b Triangles with sides 8, 7, 10 cm and 6, 7.5, 5.25 cm.

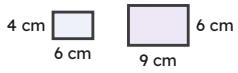
c ABC ||| DEF, AB = 8 cm, BC = 12 cm, DE = 6 cm. Find EF.

d In the same triangles, DF = 7.5 cm. Find AC.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Similar: every image side ÷ original side gives the same number · put the sides in order of size first · × or ÷ by the scale factor

5 Are the rectangles similar?



Ratios _____

Similar? _____

6 Simplify the ratio 45 : 60. Yr 8

Answer _____

7 Find x.



Scale factor = _____

x = _____

8 Similar? Sides 3, 4, 5 cm and 6, 8, 9 cm.

Ratios _____

Similar? _____

9 28 heads in 50 tosses. Observed P(head)? 12.2

Answer _____

10 ABC ||| DEF, AB = 4, DE = 10, BC = 6: EF?

Scale factor = _____

EF = _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A rectangle 8 cm long and 5 cm wide is similar to a rectangle 20 cm long. Find the width of the larger rectangle.

Answer _____

2 Two triangles are similar. The smaller has sides 6 cm, 8 cm and 10 cm. The longest side of the larger is 25 cm. Find its other two sides.

Answer _____

3 A TV screen is 80 cm wide and 45 cm high. A phone screen is 16 cm wide and 9 cm high. Are the screens similar? Show why.

Answer _____

13.4 Shadows and scale drawings

EASY

Learning intention We are learning to use shadows, scale drawings and maps to find real heights and lengths.

Success criteria
- ☐ I can find the scale factor from two shadows
- ☐ I can find a real height from its shadow
- ☐ I can use a scale like 1 : 200 to find real lengths

NOTES — Key ideas for this lesson

TEACHER COPY

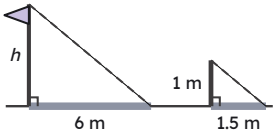
KEY TERMS

shadows: the sun makes similar triangles
scale 1 : 200: 1 cm on the plan is 200 cm
plan: a scale drawing from above
scale factor: large ÷ small, or real ÷ plan
units: make them the same first
h, d: a height, a distance

FORMULAS

Scale factor = large shadow ÷ small shadow
h = small height × scale factor
Real length = plan length × scale
100 cm = 1 m · 1000 m = 1 km

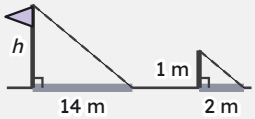
STEPS
1. change to the same units
2. find the scale factor
3. multiply the known length
4. change cm to m (or km)



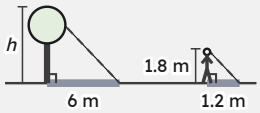
WE DO — With your teacher: copy what they write on the board.

Example 1 Find each real height or length. Change cm to m.

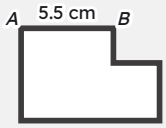
a The pole's height h



b The tree's height h

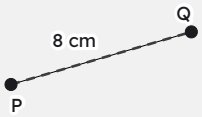


c AB, scale 1 : 100



Scale 1 : 100

d d, scale 1 : 25 000



Scale 1 : 25 000

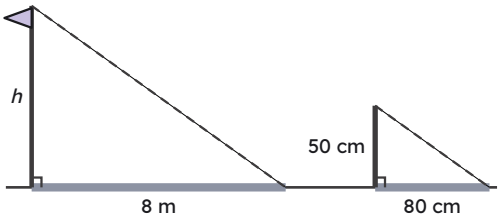
YOU DO — On your own: work down the box, one question at a time. Show your working.

1 QUESTION

REMEMBER Scale factor = large shadow ÷ small shadow · use the same units: 1 m = 100 cm · a scale 1 : 200 means real = plan × 200

QUESTION 1 Scale factor = large shadow ÷ small shadow (same units). Then find h.

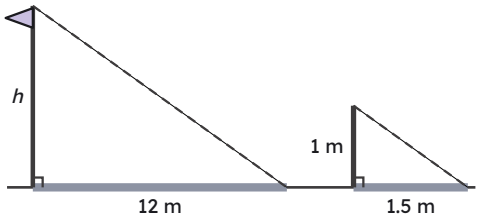
a



Scale factor = _____

h = _____

b



Scale factor = _____

h = _____

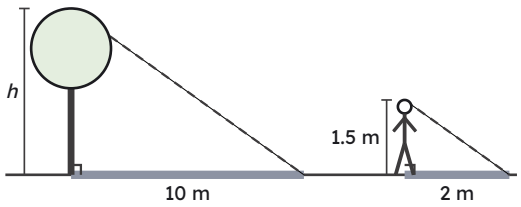
YOU DO On your own: keep going, one question at a time. Show your working. 3 QUESTIONS

REMEMBER Change every length to cm first if the units are mixed · the tall object's shadow ÷ the small one's shadow · real length = plan length × the scale

QUESTION 2

Scale factor = large shadow ÷ small shadow (same units). Then find h.

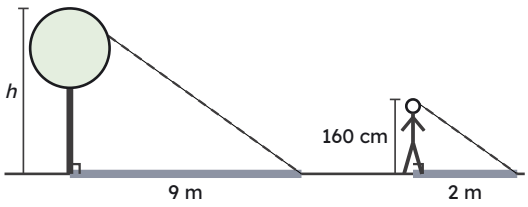
(a)



Scale factor = _____

h = _____

(b)



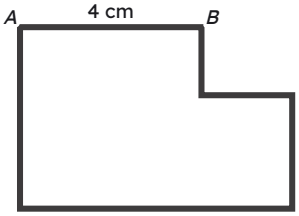
Scale factor = _____

h = _____

QUESTION 3

Scale 1 : 200 means scale factor 200. Real length = plan length × scale factor, in m.

(a)

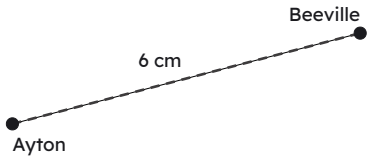


Scale 1 : 200

Scale factor = _____

AB = _____

(b)



Scale 1 : 5 000

Scale factor = _____

d = _____

QUESTION 4

Draw the two shadows. Then find the scale factor and h.

(a) A post 2 m tall has a shadow 3 m long. A building's shadow is 24 m long. Find its height h.



_____

Scale factor = _____

h = _____

(b) A girl 1.2 m tall has a shadow 80 cm long. A tree's shadow is 6 m long. Find its height h.



_____

Scale factor = _____

h = _____

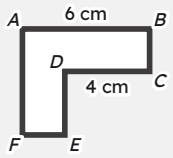
13.4 Shadows and scale drawings

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **This floor plan is drawn to a scale of 1 : 200.**

a Find the real length AB.



b Find the real length CD.

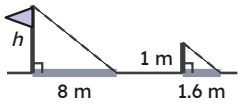
c A wall is 9 m long. How long is it on the plan?

d A door is 0.8 m wide. How wide is it on the plan?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 6 QUESTIONS

REMEMBER Real = plan × 200 · plan = real ÷ 200 · change m to cm first: 1 m = 100 cm

5 Find the pole's height h.



Scale factor = _____

h = _____

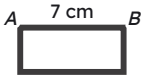
6 Divide 60 in the ratio 1 : 4. Yr 8

Answer _____

7 Map 1 : 50 000. Find d for 3 cm.

d = _____

8 Find the real length AB.



Scale 1 : 50

Scale factor = _____

AB = _____

9 Two coins: find P(at least one head). 12.4

Answer _____

10 Plan 1 : 200: how long is a 9 m wall?

Plan length _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A flagpole's shadow is 20 m long. At the same time, a stick 60 cm tall casts a shadow 150 cm long. Draw a diagram and find the height h of the flagpole.

Answer _____

2 A model plane is built to a scale of 1 : 48. The model is 35 cm long. How long is the real plane, in metres?

Answer _____

3 On a map with a scale of 1 : 25 000, two towns are 18 cm apart. Find the real distance d between the towns, in km.

Answer _____